

04-23-2025

From last:

$$Y_{it} = X_{it}^T \beta + \eta_i + e_{it}$$

$$\hat{\beta} = \left(\sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{X}_i \right)^{-1} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{Y}_i$$

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{X}_i \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{e}_i \right)$$

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{X}_i \right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{e}_i$$

$$\text{CLT: } \frac{1}{\sqrt{n}} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{e}_i \xrightarrow{d} N(0, \underline{\Omega})$$

$$\text{s.t. } \underline{\Omega} = E[\underline{X}_i^T \underline{M} \underline{e}_i \underline{e}_i^T \underline{M} \underline{X}_i] \quad \text{symmetric}$$

$$\left(\frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{X}_i \right)^{-1} \xrightarrow{p} E[\underline{x}_i^T \underline{M} \underline{x}_i]$$

$$\text{CMT: } \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \underline{V})$$

$$\text{s.t. } \underline{V} = E[\underline{X}^T \underline{M} \underline{X}]^{-1} \underline{\Omega} E[\underline{X}^T \underline{M} \underline{X}]^{-1}$$

$$\hat{\underline{V}} = \left(\frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{X}_i \right)^{-1} \hat{\underline{\Omega}} \left(\frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \underline{X}_i \right)^{-1}$$

$$\text{s.t. } \hat{\underline{\Omega}} = \frac{1}{n} \sum_{i=1}^n \underline{X}_i^T \underline{M} \hat{\underline{e}}_i \hat{\underline{e}}_i^T \underline{M} \underline{X}_i$$

Reg under TE heterogeneity not on final exam
but covariates is as simple modification of
earlier sections on reg. adjustment

GMM:

$\theta_{k \times 1}$ is variable of interest w/ $E[g(X, Y; \theta)] = 0$
moment condition $\nearrow$ $l \times 1$

case in notes: $l > k$ (more moments than θ)

We've done:

$$Y = X^T \beta + e \Rightarrow \text{linear proj} \Rightarrow \beta = E[XX^T]^{-1} E[XY] \\ \Rightarrow E[Xe] = 0 \text{ followed}$$

View as structural model:

$$Y = X^T \beta + e \Rightarrow \text{is } E[Xe] = 0?$$

$$Y = X^T \beta + \underbrace{w^T \gamma + u}_e \Rightarrow E[Xe] \neq 0 \text{ as } w \text{ unobserved}$$

$\hookrightarrow$ OVB will be $\Delta \neq \beta$

$\hookrightarrow$ need instrument to observe β .

Going to think of structural going forward:

case 1: $E[Xe] = 0 \Rightarrow X$ is exog.

$\hookrightarrow E[Xe]$ is a moment condition

$$= E[X(Y - X^T \beta)] = 0 = g(X, Y; \beta) \text{ is } k \times 1$$

$$\Rightarrow E[XY] = E[XX^T] \beta \Rightarrow \beta = E[XX^T]^{-1} E[XY]$$

case 2: $E[Xe] \neq 0$

ex: x_1 - educ; x_2 - age; x_3 - int

$l < k$ $E[X_1 e] \neq 0$ (X_1 is endog.) β is underidentified
 $E[X_2 e] = 0$ or fails order condition
 $E[X_3 e] = 0$

GMM:

case 2: $E[Xe] \neq 0$

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \text{ s.t. } \begin{matrix} X_1 \text{ are endog.} \\ X_2 \text{ are exog.} \end{matrix}$$

Find another $k \times 1$ vector of random vars, Z , satisfying $E[Ze] = 0$

$$Z = \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}_{k \times 1} \text{ s.t. } Z_2 = X_2$$

Z_1 is an instrumental variable (iv)

$$E[Z(Y - X^T \beta)] \Rightarrow E[ZY] = E[ZX^T] \beta \\ \Rightarrow \beta = \underbrace{E[ZX^T]^{-1}}_{\text{works when full rank } k \text{ or relevance condition, } E[Z, X_1] \neq 0} E[ZY]$$

$\hookrightarrow$ works when full rank k
or relevance condition,
 $E[Z, X_1] \neq 0$

Over-identification: $Z_{l \times 1}$ s.t. $l > k$

$E[ZX^T]_{l \times k} \Rightarrow$ not invertible

$\hookrightarrow$ can truncate but wasting data

$E[ZX^T] \beta = E[ZY]$ still works, hard to proceed

$$O = \left(\frac{1}{n} \sum_{i=1}^n Z_i X_i^T \right) \hat{\beta} - \frac{1}{n} \sum_{i=1}^n Z_i y_i \quad (*)$$

$\hookrightarrow$ won't be able to find $\hat{\beta}$ exactly, approximate

Idea of GMM: make RHS of $*$ as close to 0 as poss.

$$\hat{\beta} = \arg \min_b \left(\underline{z}^T y - \underline{z}^T x b \right)^T \hat{W} \left(\underline{z}^T y - \underline{z}^T x b \right)$$

$\hat{W}$ is weighting matrix

General Version:

$$\hat{\theta} = \arg\min_c \left(\frac{1}{n} \sum_{i=1}^n g(y_i, x_i; c) \right)^T \hat{\omega} \left(\frac{1}{n} \sum_{i=1}^n g(y_i, x_i; c) \right)$$

Back to overidentified:

$$\hat{\beta} = \arg\min_b y^T z \hat{\omega} z^T y - 2b^T x^T z \hat{\omega} z^T y \\ + b^T x^T z \hat{\omega} z^T x b$$

$$\text{FOC: } b: 0 = -2x^T z \hat{\omega} z^T y + 2x^T z \hat{\omega} z^T x \hat{\beta}$$

$$\Rightarrow \hat{\beta} = (x^T z \hat{\omega} z^T x)^{-1} x^T z \hat{\omega} z^T y$$